Kyler Cao

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Education

Texas A&M University College Station, TX

Bachelor of Science in Computer Science, Minor in Business | ***GPA: 3.59/4.00*** *Aug. 2023 - May 2027*

- **Relevant Coursework:** Data Structures & Algorithms, Computer Organization, Foundations of Software Engineering, Database Systems, Machine Learning

Experience

Associate Software Engineer Aug. 2026 - Present

Global Shop Solutions *Remote*

- Contributing to GSS Cloud through **CI/CD pipeline** improvements and production-server feature development
- Building an internal developer **code harness** that connects engineers to company services and tools for reusable workflows

Product & Engineering Intern May 2026 - Aug. 2026

Global Shop Solutions *The Woodlands, TX*

- Developed components for **Launchpad**, an internal developer platform using **Go, PostgreSQL, and Kubernetes** that automates declarative application deployment through GitHub and Argo CD with reconciled jobs, Keycloak authentication, and department-scoped access control, centralizing application delivery and access management
- Built an **AI-assisted multimodal diagnosis dashboard** with React, Node.js, and PostgreSQL that correlates test evidence with structured model outputs to rank causes and generate reviewable diagnoses, reducing average ticket resolution time by **70%**
- Automated **COBOL-to-C# modernization validation** across a multi-VM fleet by building a React/TypeScript/.NET compilation dashboard and paired TestArchitect regression pipeline that streams build artifacts, enables direct behavioral comparison, increasing weekly modernization throughput from **2.5% to 10%**

Internal Tools Developer Oct. 2025 - May 2026

Texas A&M TechHub *College Station, TX*

- Created a full-stack delivery management system with **Flask, React, and MySQL** that processes **150+ monthly orders** and maintains auditable order histories
- Delivered **8+ internal tools** spanning inventory automation, documentation, and asset lifecycle tracking for a **1,000+ device** campus fleet
- Deployed **1,000+ university devices** and resolved **200+ faculty and staff support cases**, cutting average imaging time by **15%**

Teaching Assistant Aug. 2024 - May 2025

Texas A&M University *College Station, TX*

- Taught foundational **Python** to **120+ students** through labs, office hours, and weekly problem-solving reviews
- Evaluated **400+ assignments and exams** with a **24-hour turnaround**, providing targeted feedback and creating exam-review materials

Projects

CodeHawk | *Go, Python, LangGraph, Redis Streams, pgvector* | Private Project June 2026 - Present

- Built the core of a tiered GitHub PR review system with a **Go API** for signed webhook ingestion, diff parsing, Redis Streams job dispatch, and PR publication, plus a **Python/LangGraph worker** for agentic review orchestration
- Implemented durable review lifecycle controls—including transactional outbox, immutable archives, idempotency, and retry/dead-letter handling—and documented a reproducible Docker Compose workflow for local verification

Seer | *Rust, ratatui, crossterm, iroh, portable-pty* | Private Project Sept. 2026 - Present

- Built a multi-crate **Rust** terminal runtime for coding agents with a **ratatui/crossterm** TUI, persistent workspace and pane state, and local PTY runtimes
- Implemented brokered multi-user sessions with credentialed client attachments, end-to-end encrypted remote connections via **iroh**, and integration tests for handshakes, forwarding, detach, and runtime lifecycle

Technical Skills

Languages: Rust, Python, Go, TypeScript/JavaScript, C#, SQL, Java, C++, COBOL
Frameworks & Libraries: React, .NET, Flask, Node.js, LangGraph, ratatui, crossterm, portable-pty, pandas, scikit-learn
Tools & Platforms: Git, Docker, Kubernetes, Argo CD, Linux, PostgreSQL, MySQL, Redis, iroh, Slurm